

Physics of Semiconductor Devices

ECE210: PSD



INDRAPRASTHA INSTITUTE *of*
INFORMATION TECHNOLOGY
DELHI

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(week 09)



Non-ideality in Schottky junction



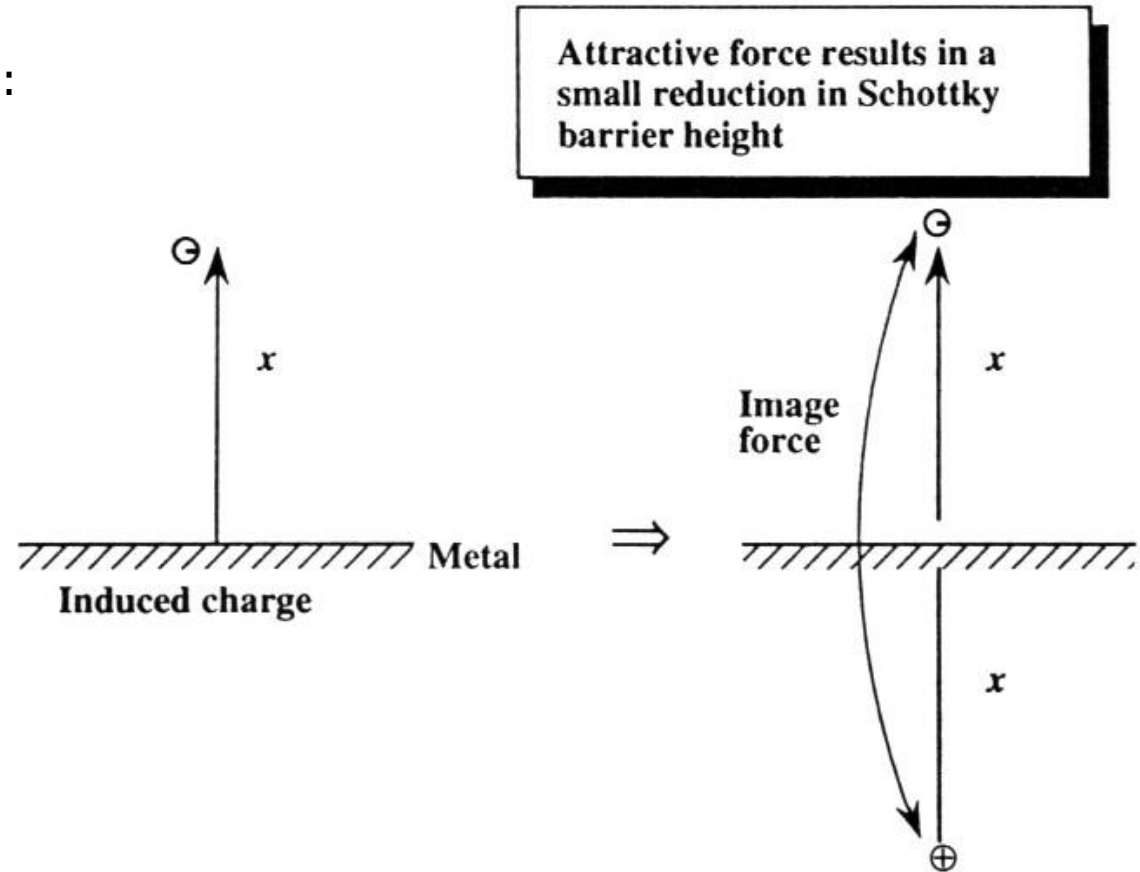
Modifications to the Barrier Potential (Image-Force Lowering)

Force between the two electrons :

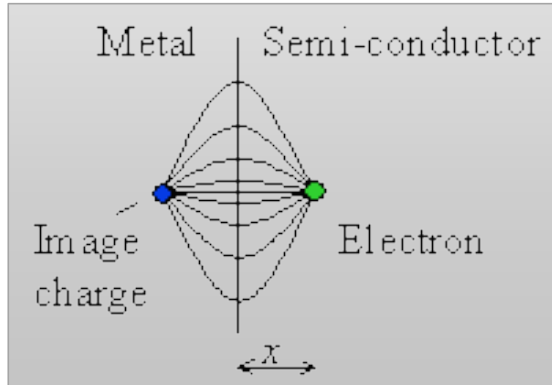
$$F = \frac{q \times (-q)}{4\pi\epsilon_0 (2x)^2} \quad [\text{Coulomb's law}]$$

Potential energy for an electron :

$$V_{IF}(x) = \int_{\infty}^x F dx = \frac{q^2}{16\pi\epsilon_0 x}$$



Modifications to the Barrier Potential (Image-Force Lowering)



Under an electric field E :

$$V(x) = \frac{q^2}{16\pi\epsilon_0 x} + q\vec{E}x$$

The potential energy due to the distributed charge of the ionized donors and a single electron reaches its maximum value at

$$x_{\max} = \sqrt{\frac{1}{16\pi\epsilon_s E_{\max}}}$$

and the barrier height reduction is $\Delta\Phi_B = \sqrt{\frac{qE_{\max}}{4\pi\epsilon_s}}$

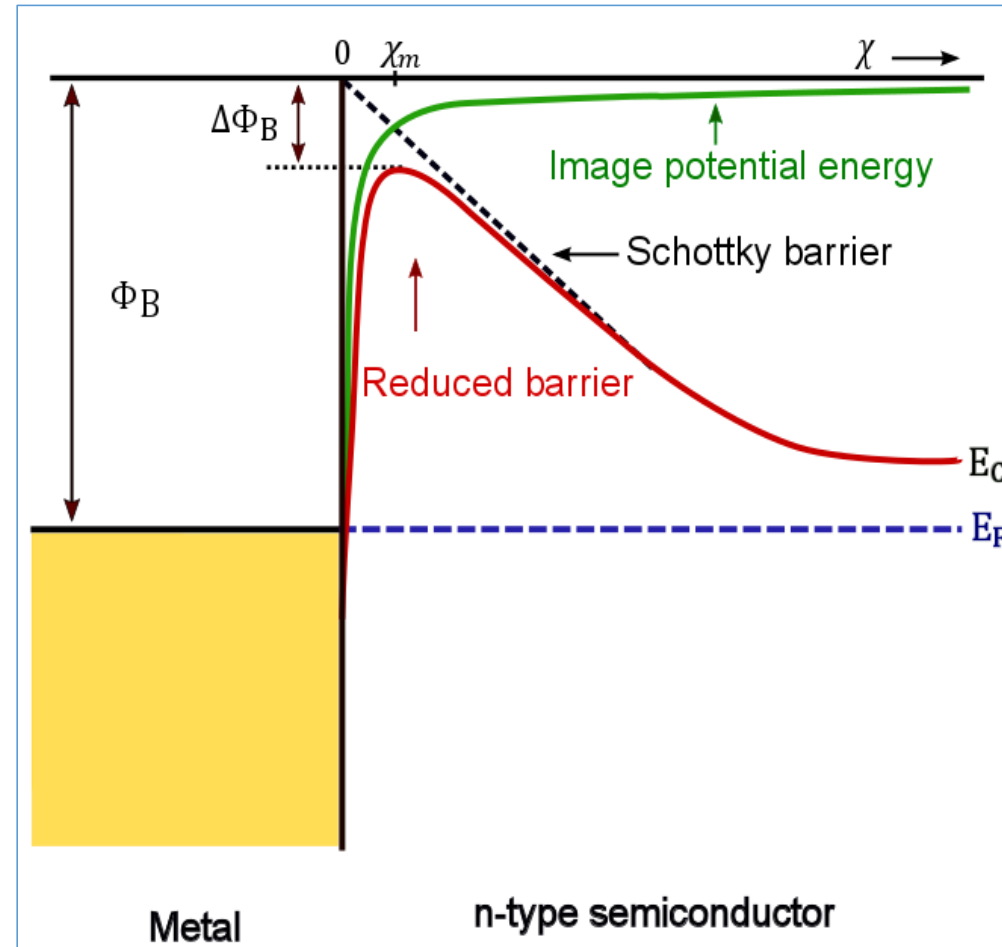
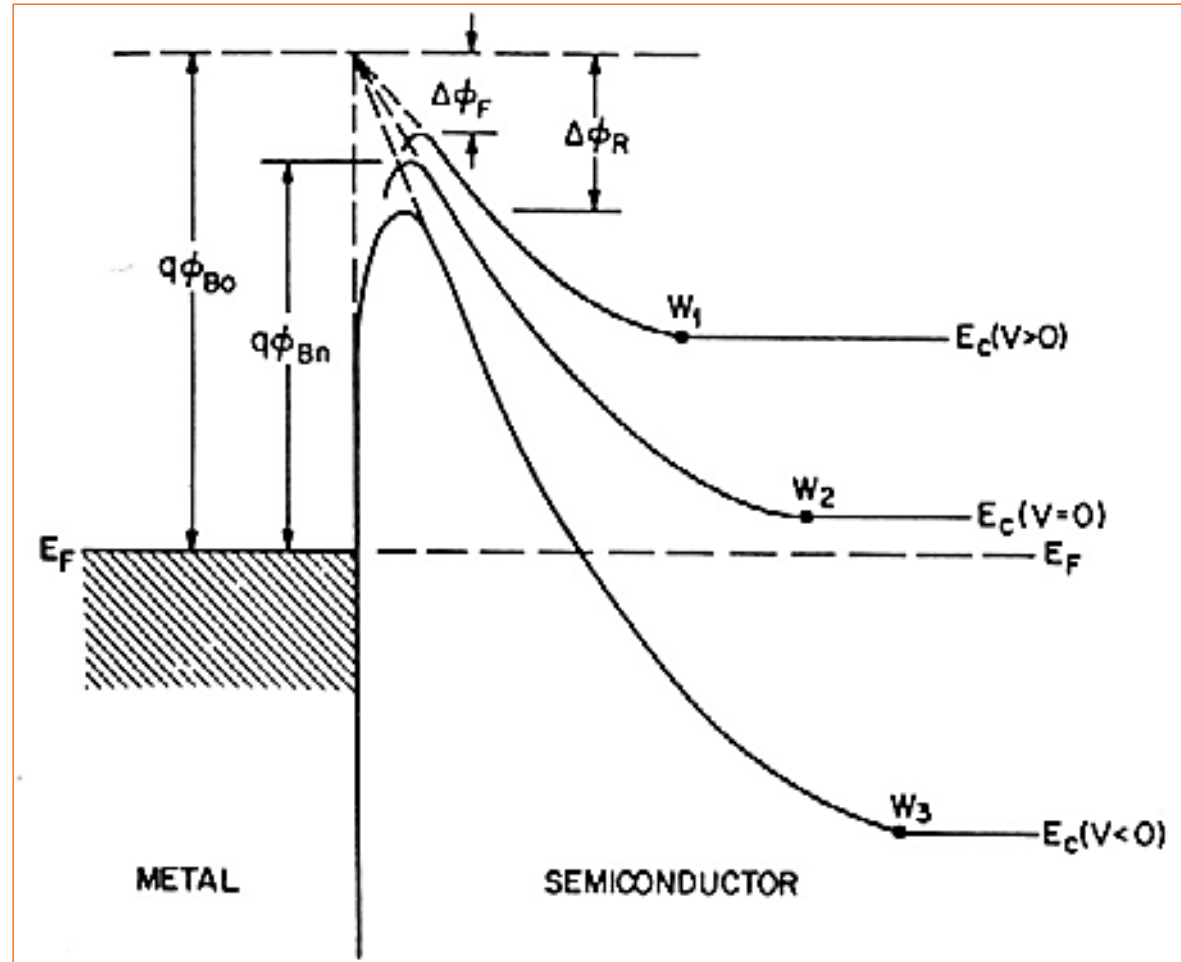
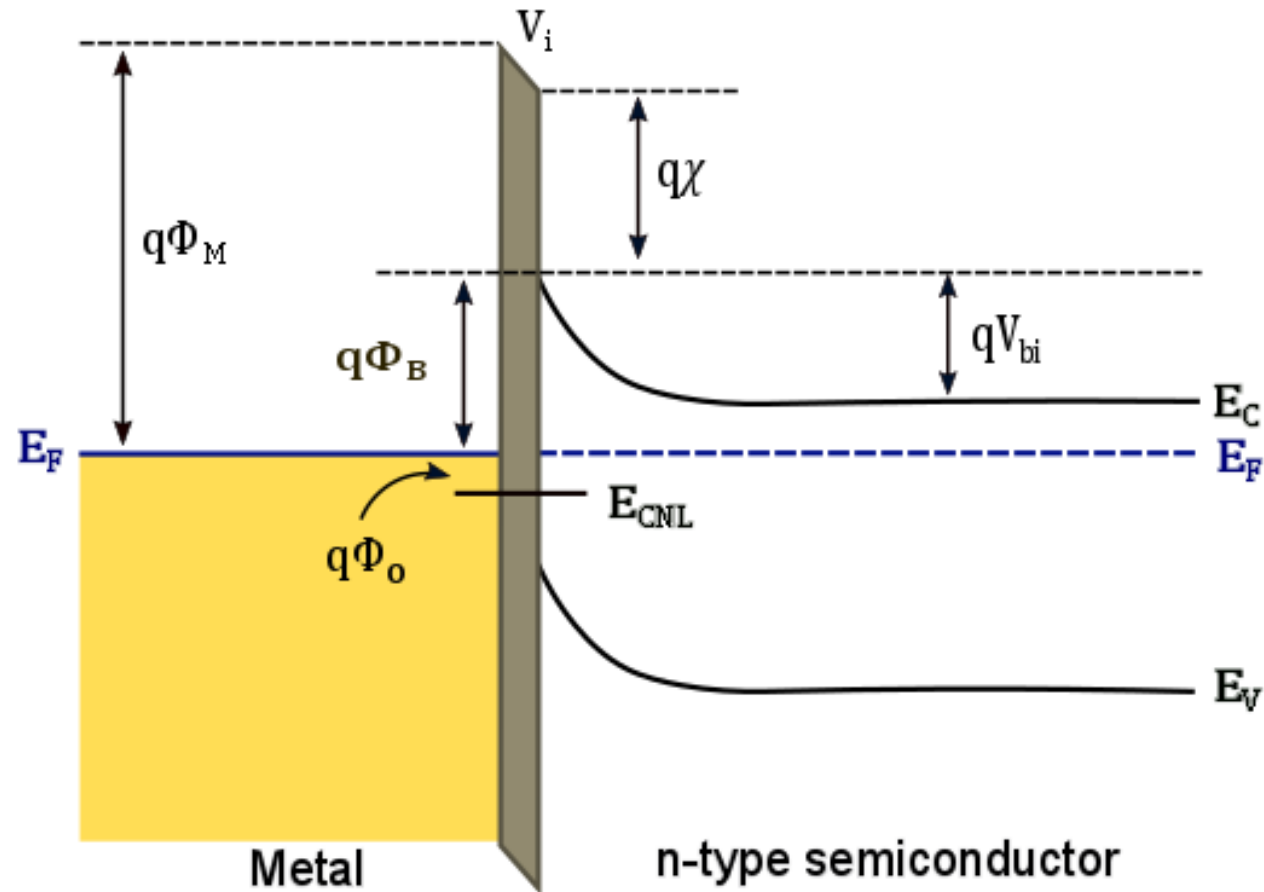


Image-Force Lowering under bias



Modifications to the Barrier Potential (Interfacial Layers)

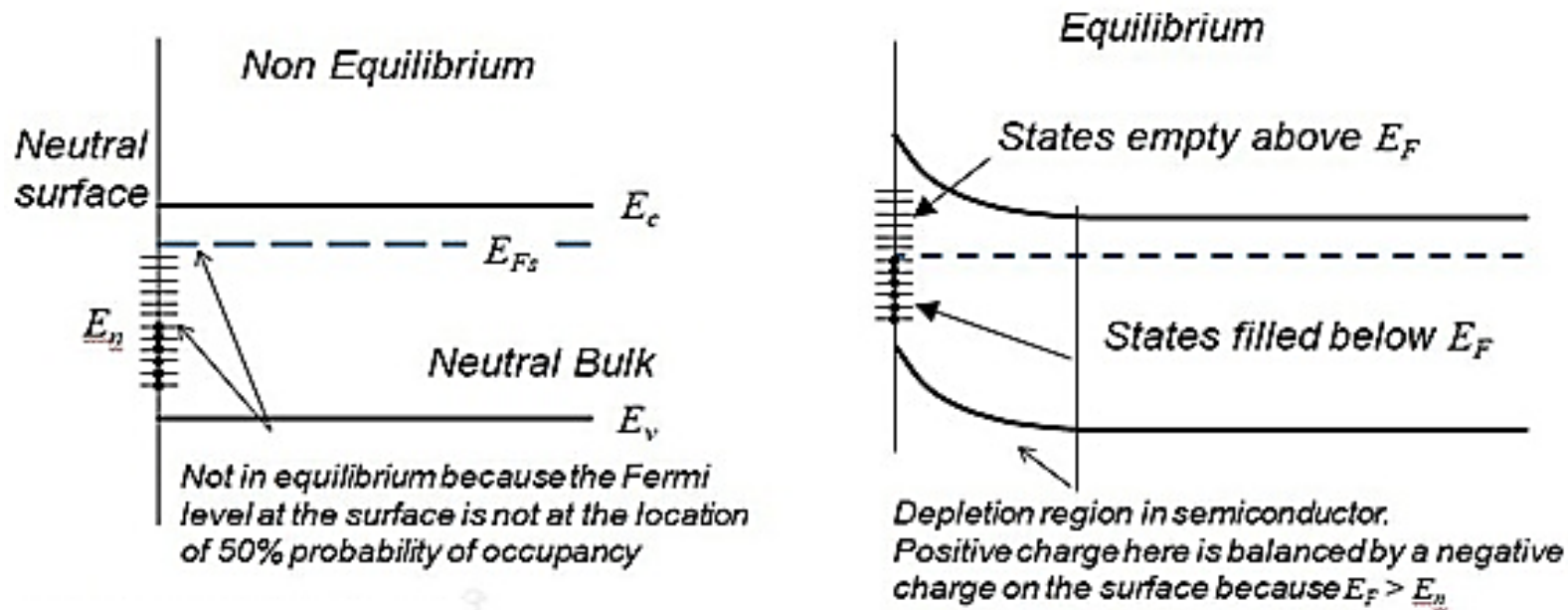


Interface states: Fermi level pinning



Surface States and Fermi-Level Pinning

- Because the surface does not have the same periodicity as the bulk there may be fixed electron states in the forbidden band. Lower energy states may be filled, higher energy states may be empty.



Metal Induced Gap States (MIGS) → another origin of interface states that results in "metallic" screening → Fermi level pinning

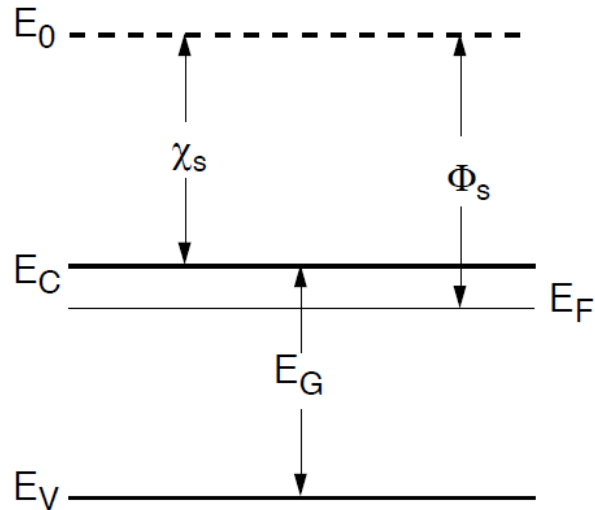
Semiconductor – semiconductor junctions



semiconductor

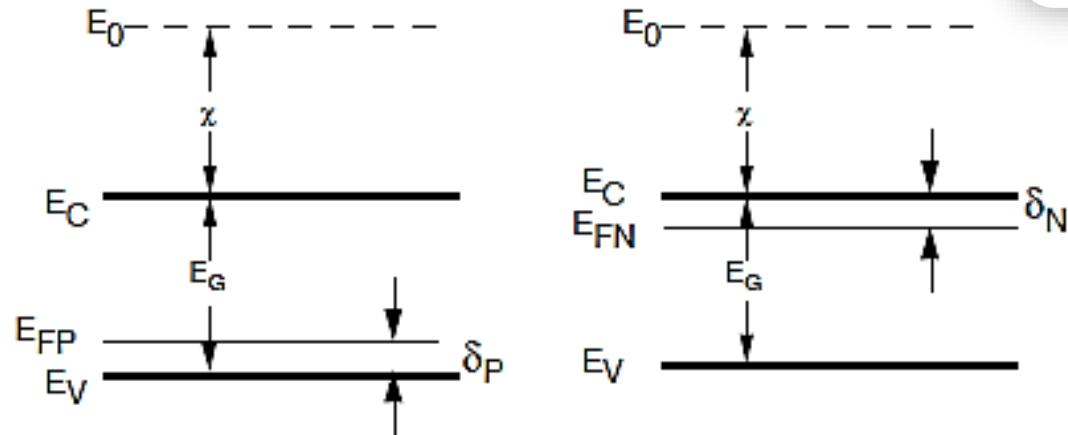
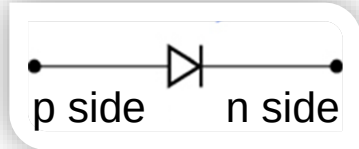
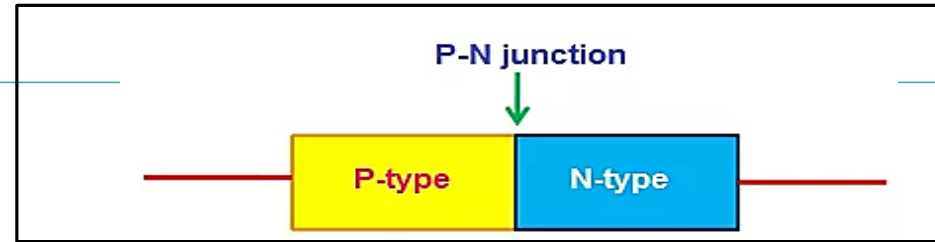


the energy band diagram for a uniformly doped, compositionally uniform semiconductor



$$E_C = E_0 - \chi_s$$

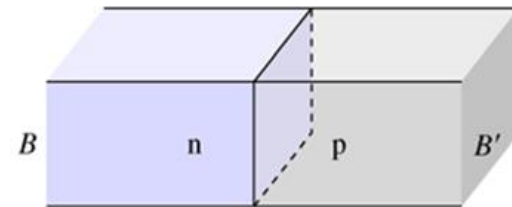
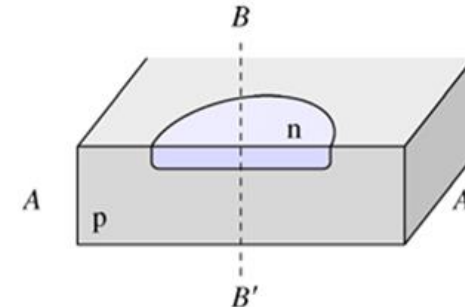
$$E_V = E_C - E_G = E_0 - \chi_s - E_G$$



Prototype of p-n homojunctions

Example:

- start with a p-type substrate
- Implant phosphorus atoms (donors) in certain region
- Junction occurs along B-B' cross section



Semiconductor – semiconductor junctions

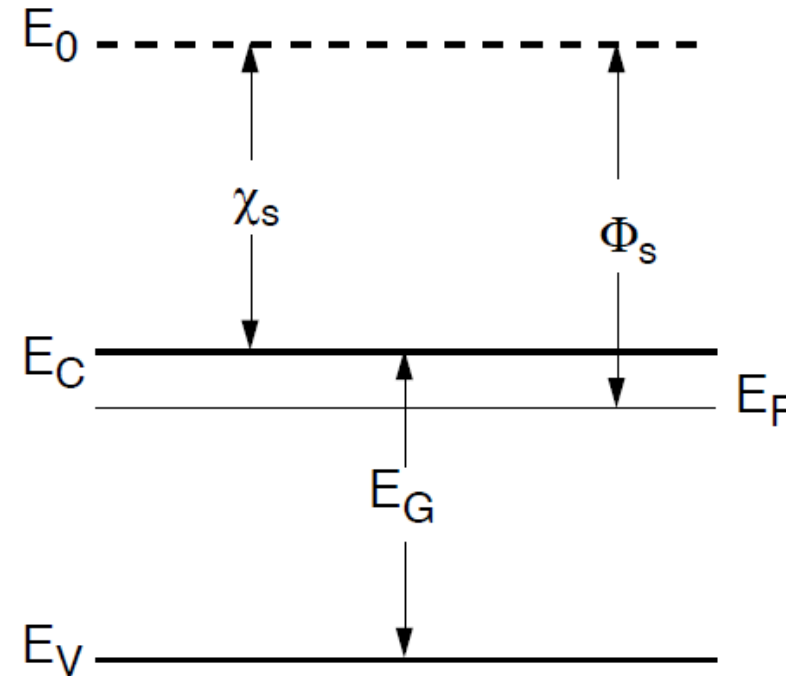


semiconductor



the energy band diagram for a uniformly doped, compositionally uniform semiconductor

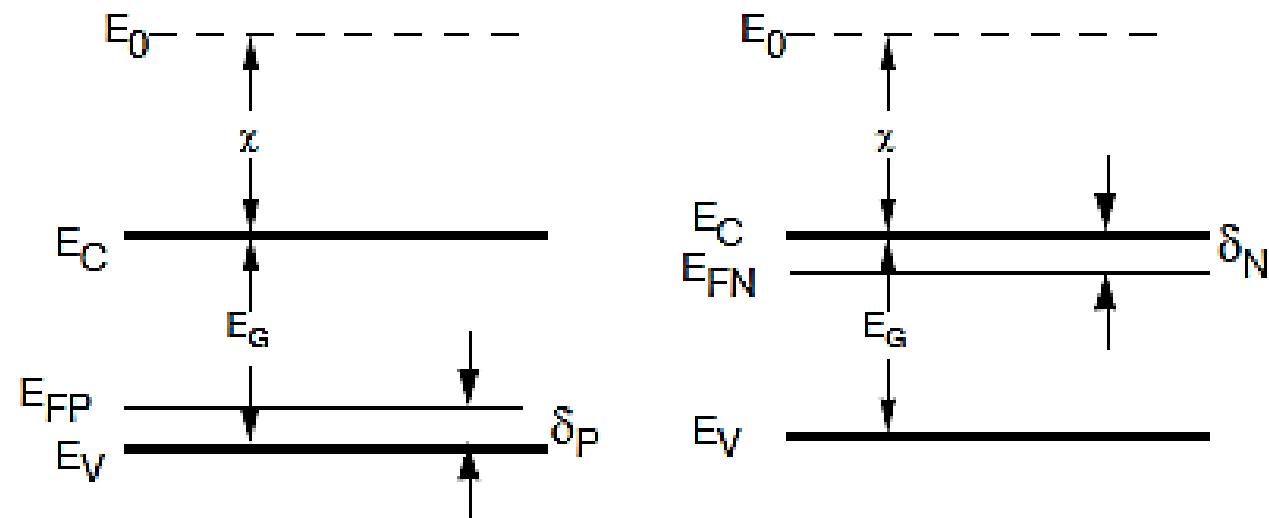
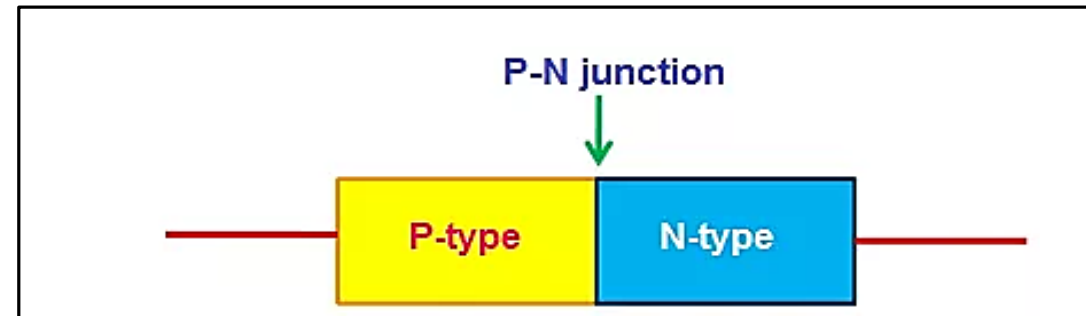
The location of the conduction band is measured with respect to a reference energy level, E_0 , the field-free vacuum level.



$$E_C = E_0 - \chi_s$$

$$E_V = E_C - E_G = E_0 - \chi_s - E_G$$

Homojunctions



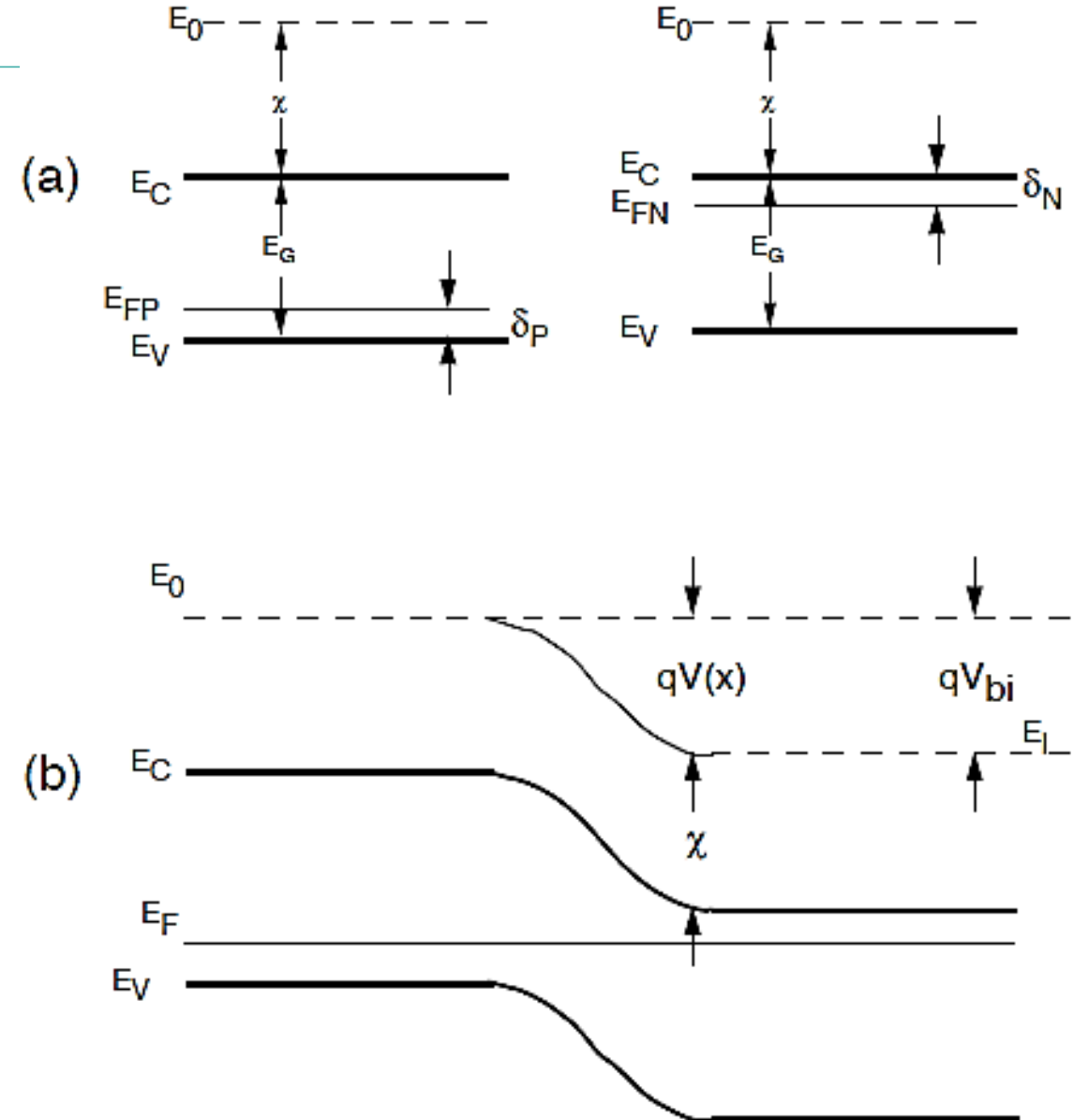
Homojunctions



Illustration of the formation of a pn junction.

(a) the isolated semiconductors are characterized by separate Fermi levels, and

(b) after placing them in contact, electrons flow from the region with the higher Fermi level to the region with the lower Fermi level. As a result of this flow, a space-charge layer builds up and a built-in potential results. In equilibrium, the final structure is characterized by a single Fermi level. Note that E_0 is lowered to $E_i(x)$, the local vacuum level



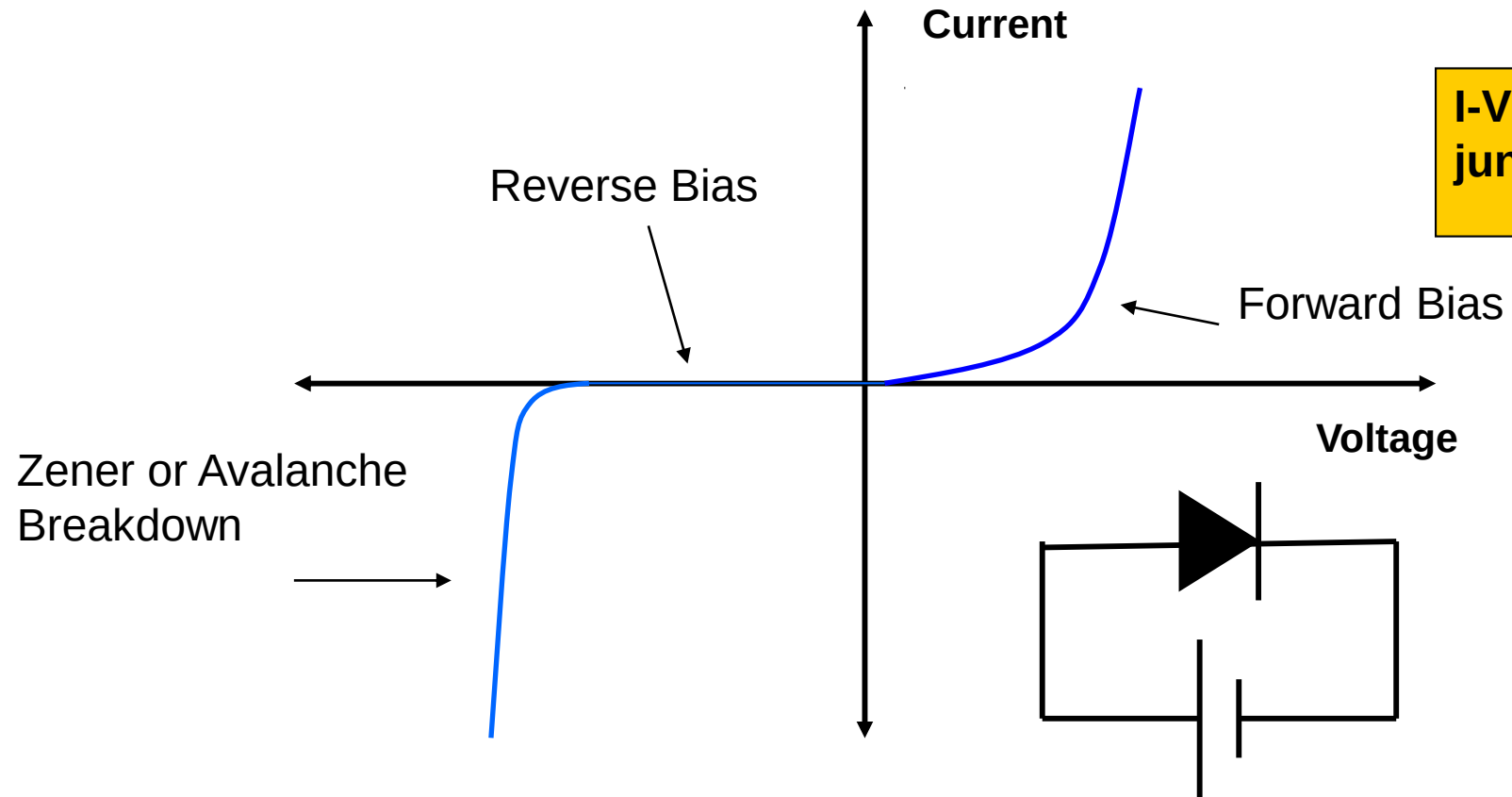
$$qV_{bi} = \Phi_P - \Phi_N$$

where

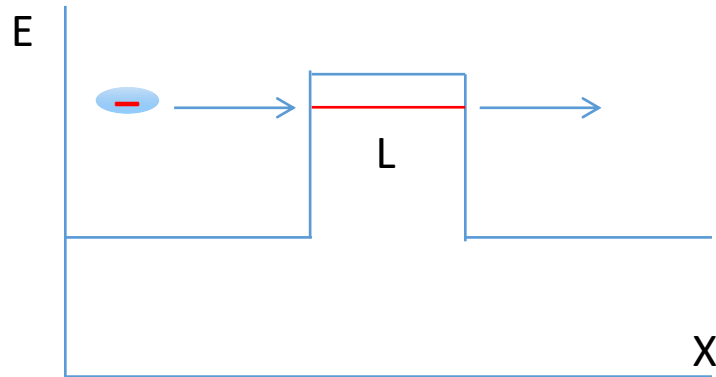
$$\Phi_P = \chi_P + E_{GP} - \delta_P$$

$$\Phi_N = \chi_N + \delta_N$$

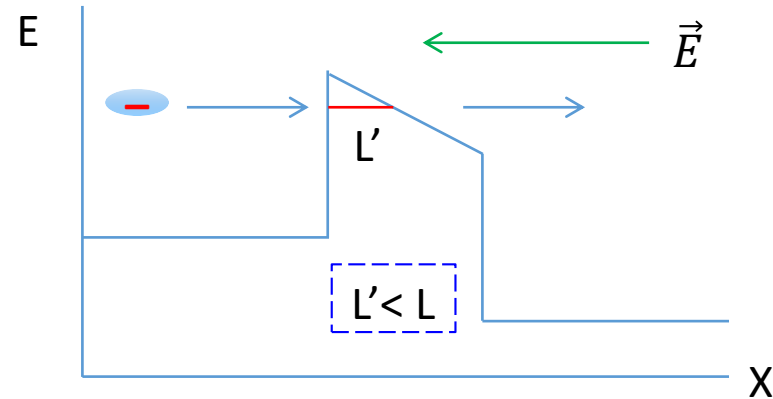
pn diode characteristics:



Tunneling (one kind of transport mechanism)



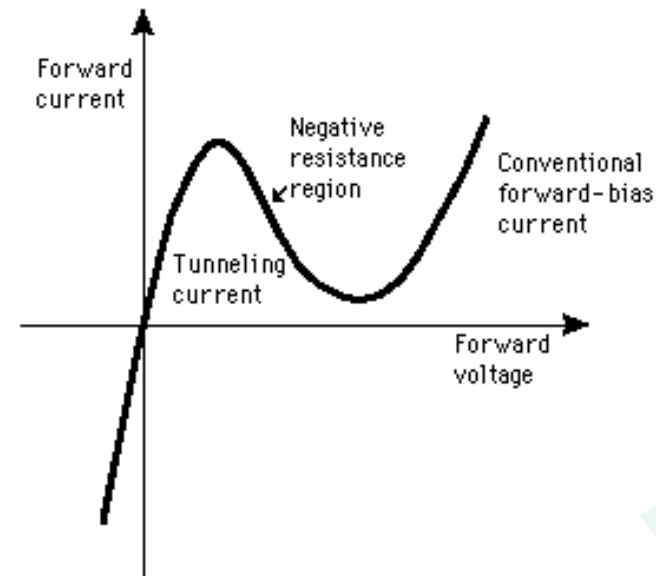
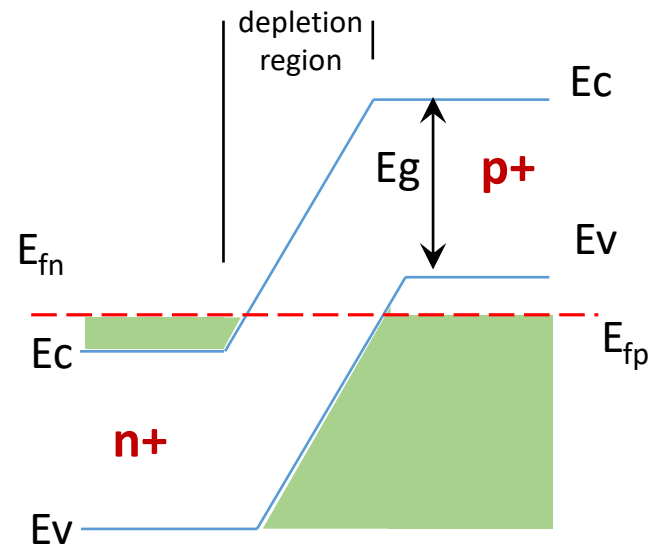
tunneling barrier in its ground state



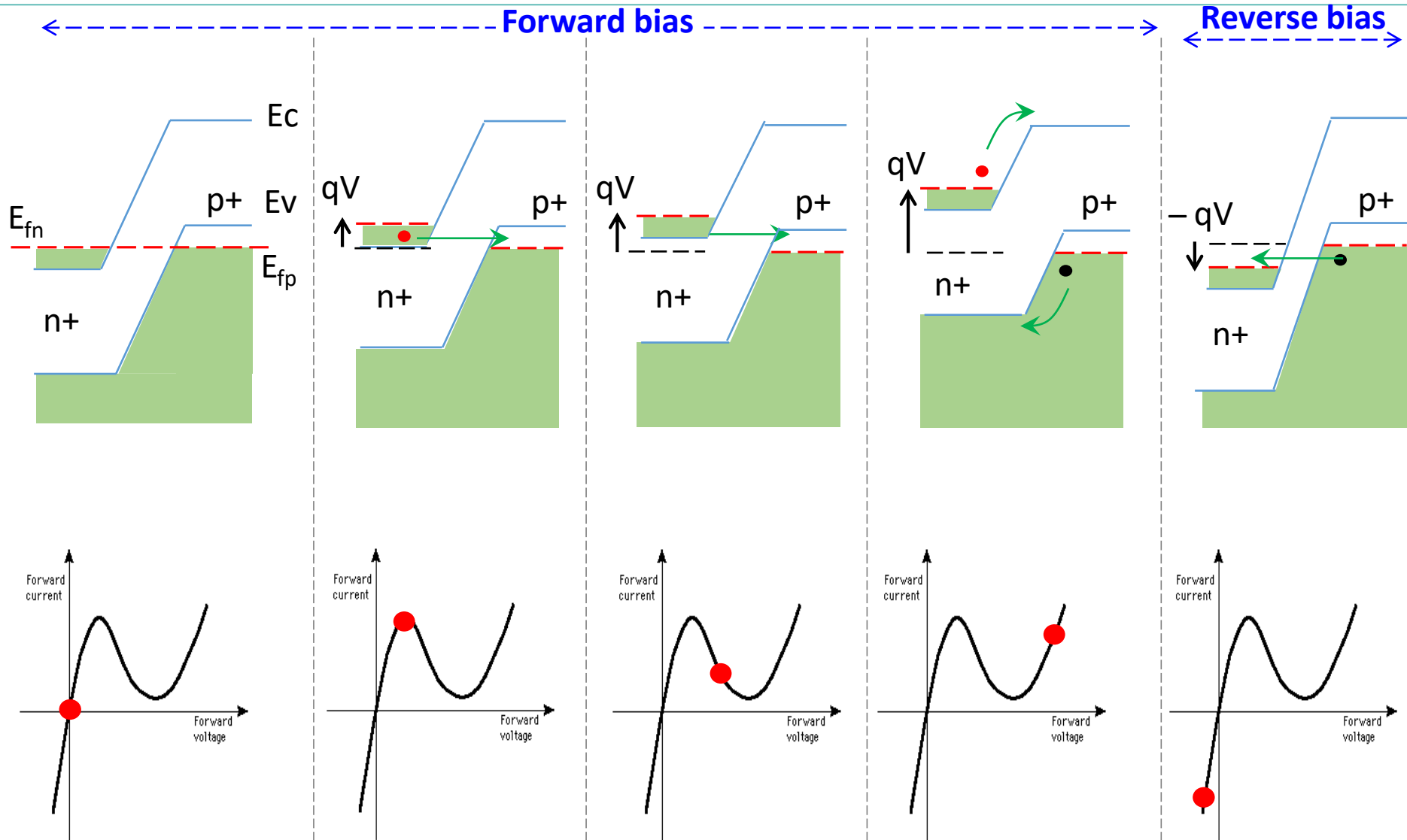
tunneling element with applied voltage

Tunneling is only possible if a free band is available on the other side of the barrier

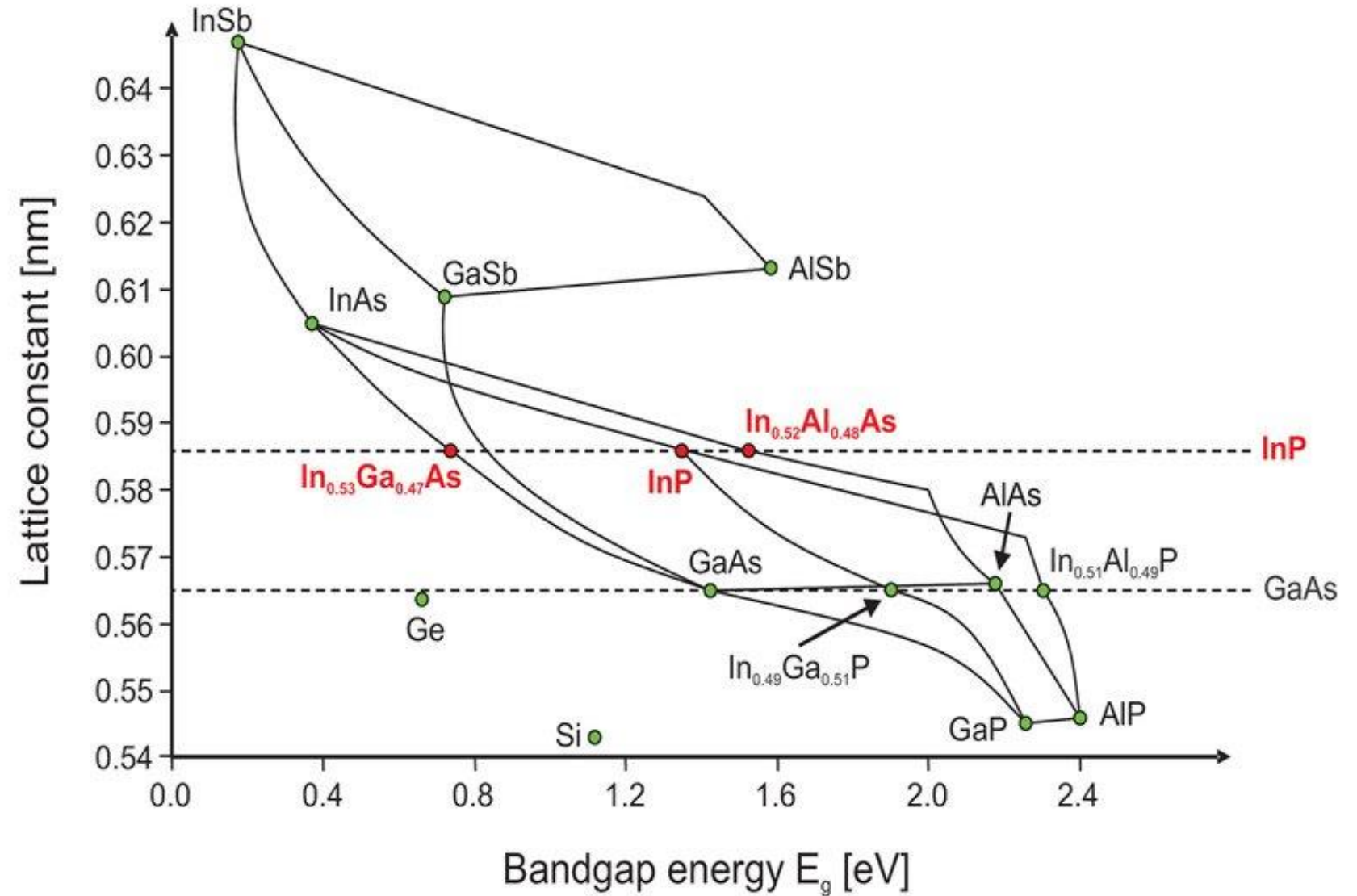
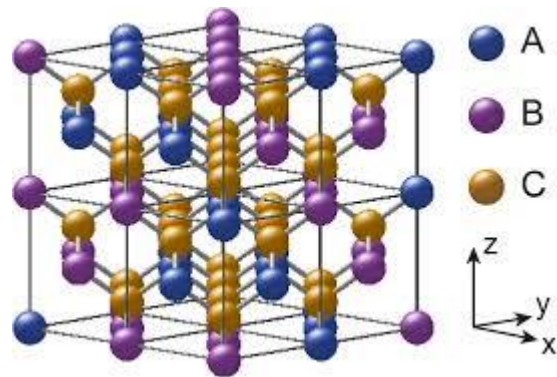
Tunnel diode (invented by Leo Esaki in 1957)



Tunnel diode under bias



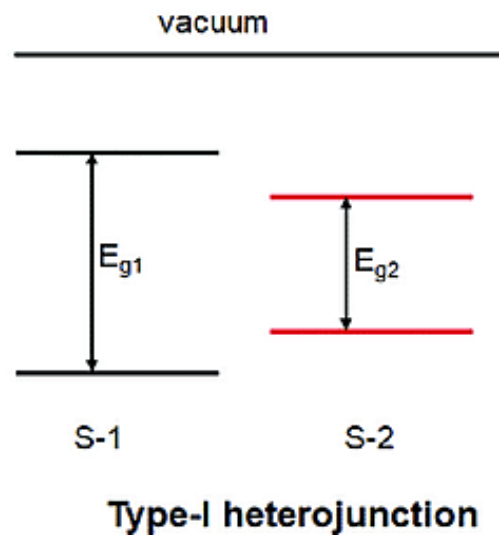
Band structure engineering



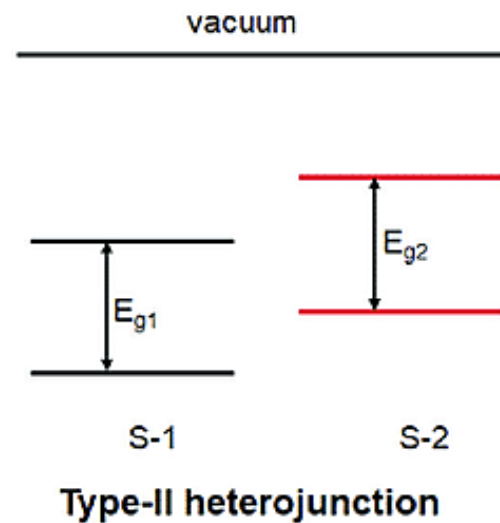
Heterojunctions



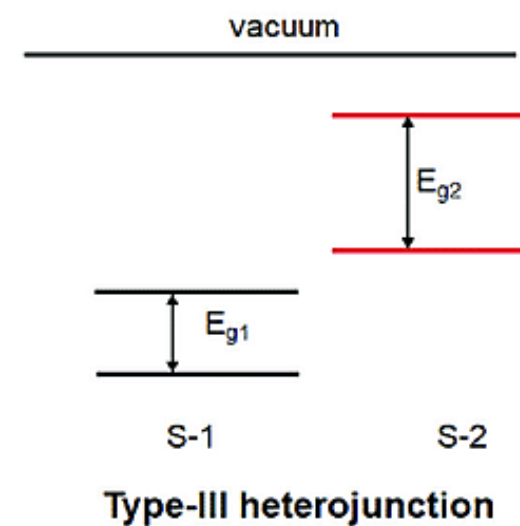
To draw energy band diagrams for devices with a position-dependent alloy composition (so-called *heterostructure devices*), it is essential to know more than simply how the band gap varies with position - we must also know how the bands line up at compositional junctions.



Straddling Gap



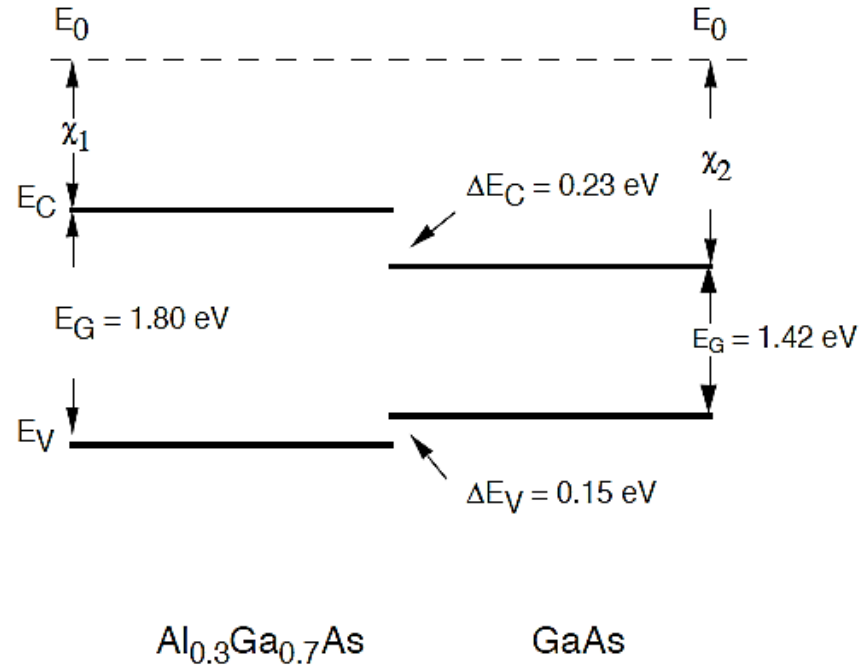
Staggered Gap



Broken Gap

(type II misaligned heterojunction)

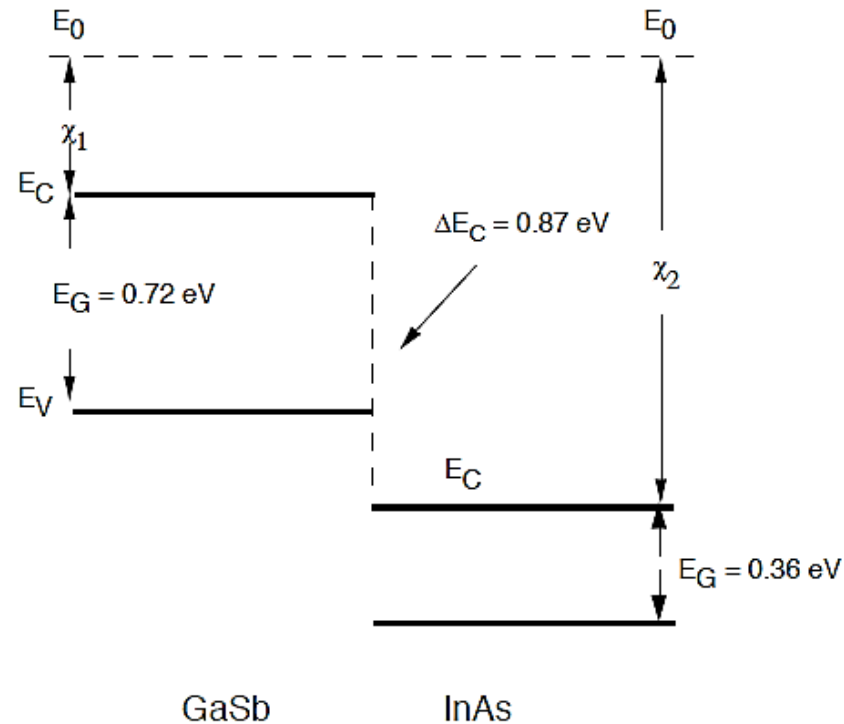
Heterojunctions



$$\Delta E_C = \chi_2 - \chi_1$$

$$\Delta E_G = \Delta E_C + \Delta E_V$$

Observed band alignments for $\text{Al}_x\text{Ga}_{1-x}\text{As}$ for $x = 0.3$ and $x = 0.0$. These are known as type I heterojunctions.



Observed band alignments for GaSb and InAs . These are known as Type III heterojunctions, or as Type II misaligned heterojunctions

Examples of type II heterojunctions include the $\text{In}_x\text{Ga}_{1-x}\text{As} / \text{Ga}_x\text{Sb}_{1-x}\text{As}$ and $\text{Al}_x\text{In}_{1-x}\text{As} / \text{InP}$ systems

Heterojunctions



$\text{Al}_{0.3}\text{Ga}_{0.7}\text{As}$: GaAs / N-p heterojunction

(By convention, the capital letter identifies the semiconductor with the larger bandgap)

(a) before contact. (b) after contact.

When placed in contact, electrons move from the semiconductor with the higher Fermi-level to the other, and an electric field is produced to balance this transfer. The built-in potential is simply the difference in workfunctions of the isolated, bulk semiconductors as

$$qV_{bi} = \Phi_P - \Phi_N$$

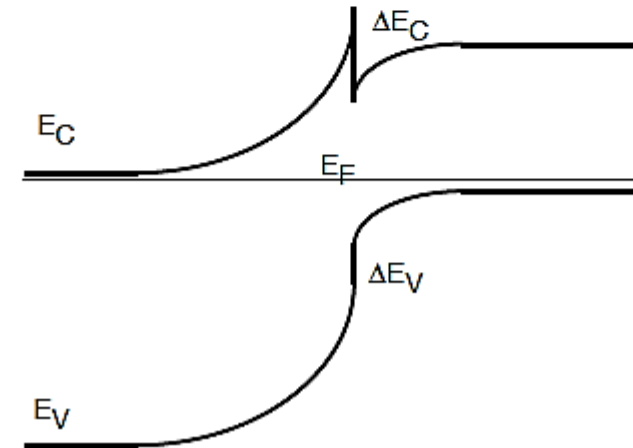
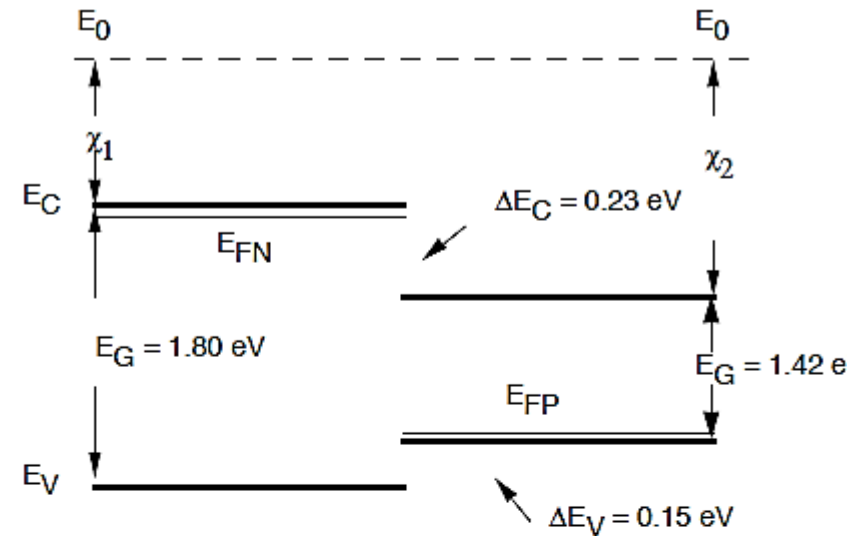
where

$$\Phi_P = \chi_P + E_{GP} - \delta_P \quad (b)$$

$$\Phi_N = \chi_N + \delta_N$$

when $\Delta E_c = 0$; Graded Heterojunctions

(a)

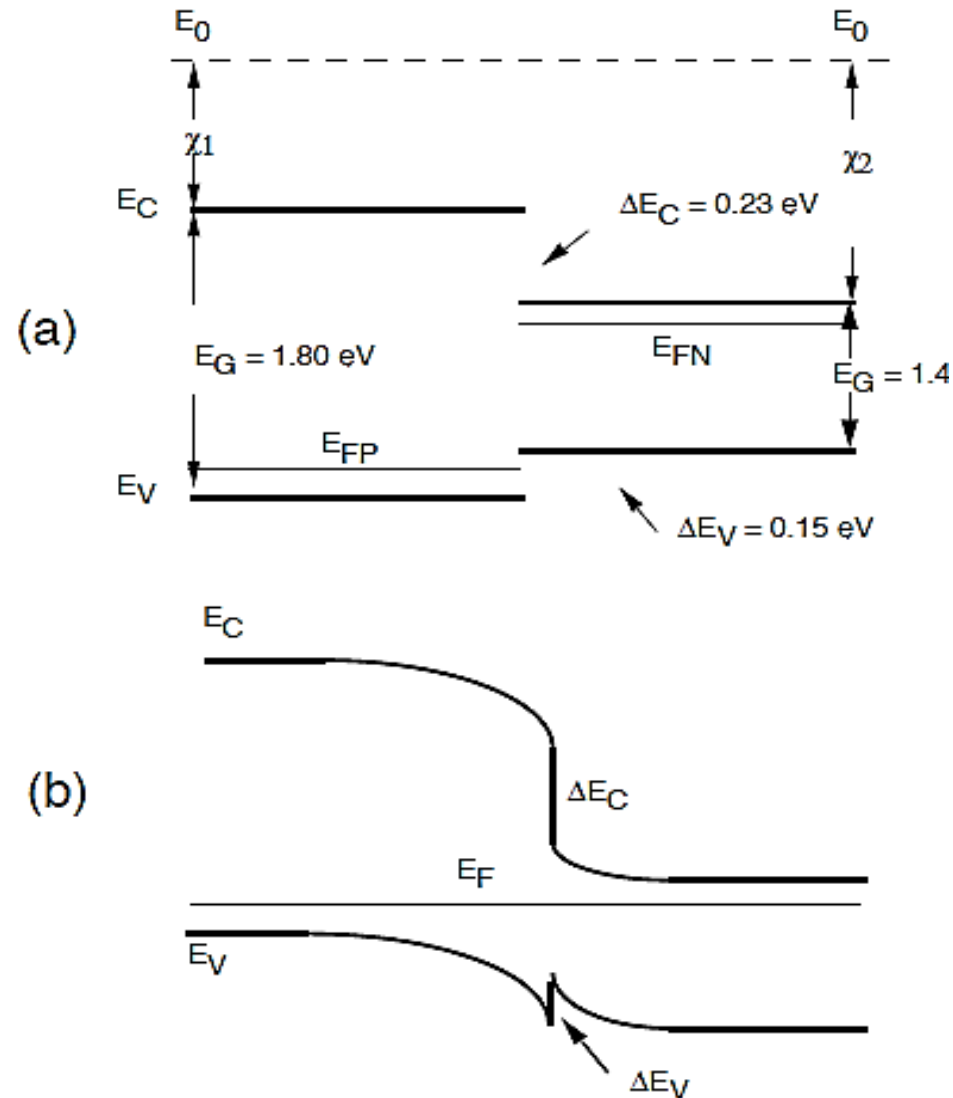


Heterojunctions



$\text{Al}_{0.3}\text{Ga}_{0.7}\text{As}$: GaAs/ P-n heterojunction

(a) before contact. (b) after contact.

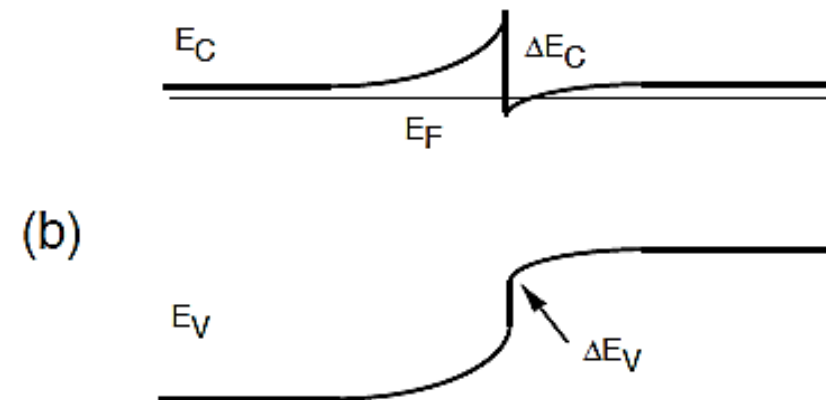
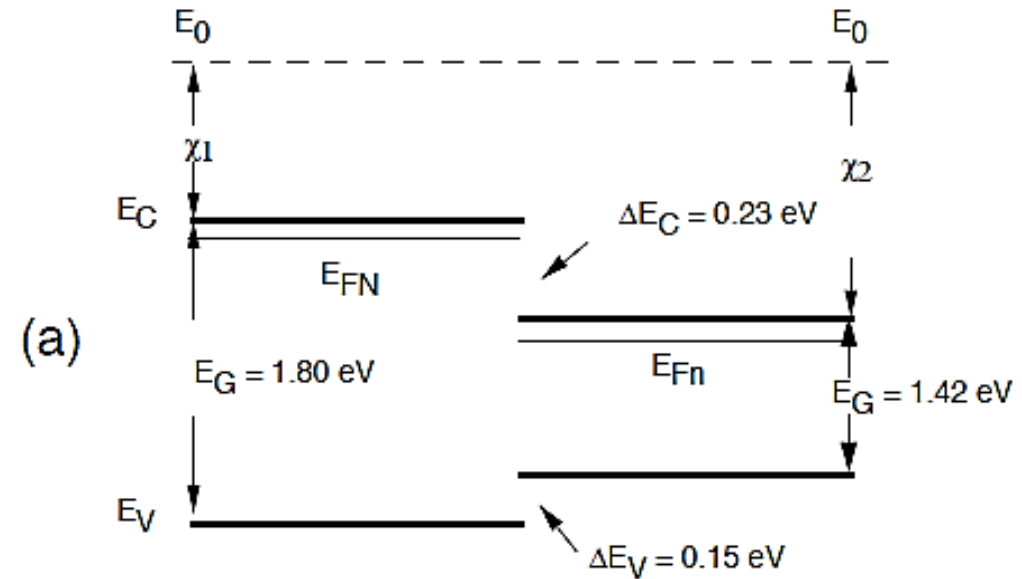


Heterojunctions



$\text{Al}_{0.3}\text{Ga}_{0.7}\text{As}$: GaAs/
N-n heterojunction.

(a) before contact. (b) after contact



Heterojunctions

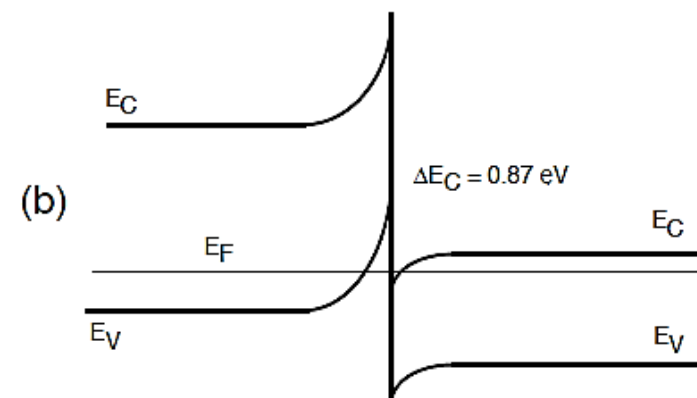
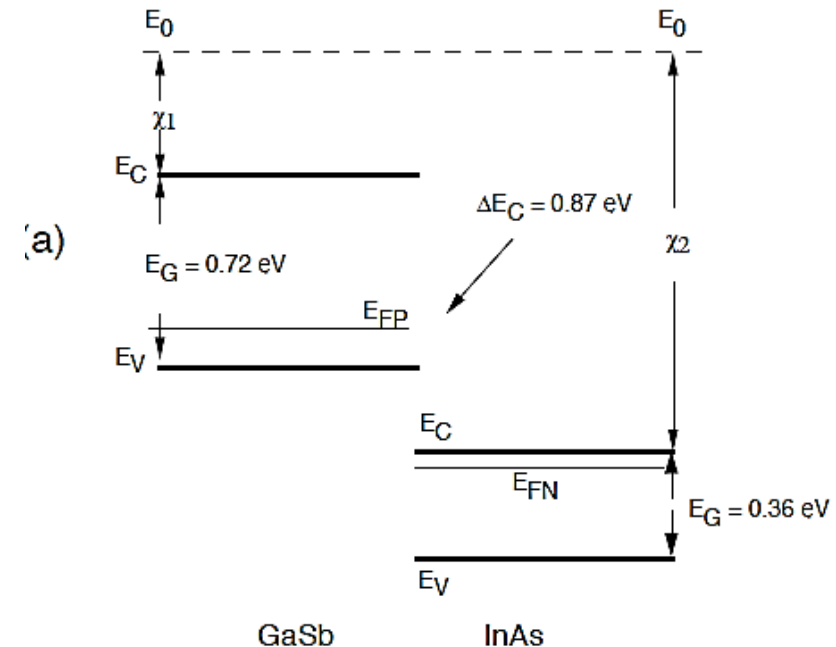


GaSb : InAs/ P-n heterojunction

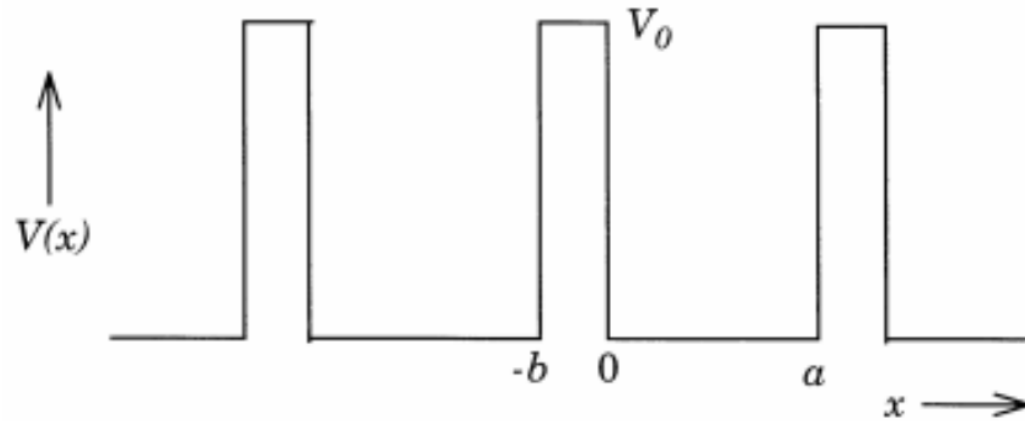
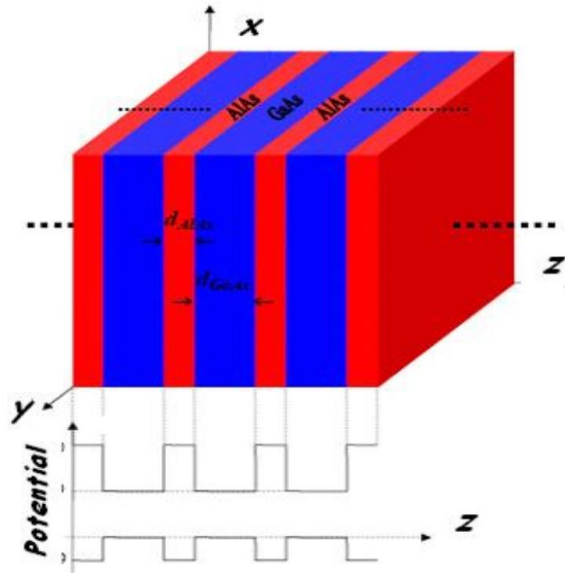
(a) before contact. (b) after contact

we have an accumulation of electrons on the n-InAs side and an accumulation holes on the P-GaSb side.

Because the conduction band on the InAs side of the interface lies lower than the valence band on the GaSb side, this interface is sometimes referred to as “semimetallic.”



Superlattice



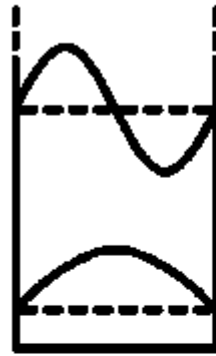
the periodic potential with period $d = a + b$. It has the form

$$V(x) = \begin{cases} 0 & -b < x < 0 \\ V_0 & 0 < x < a \end{cases}$$

miniband

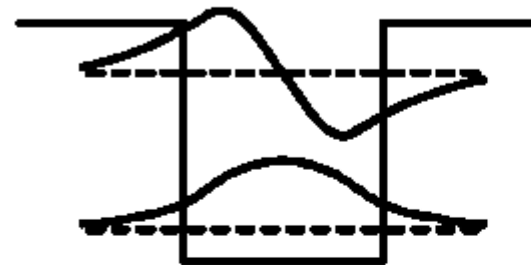


"infinite" quantum well



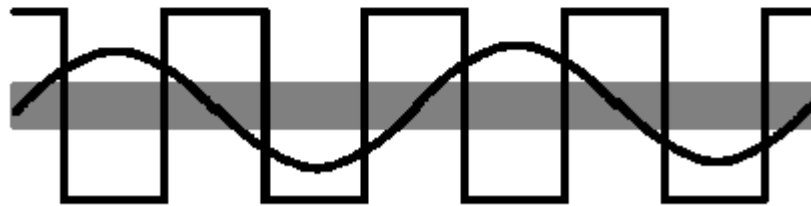
"particle in a box"

finite quantum well



particle in a box +
"tunneling"
penetration

superlattice



wells so close that
wavefunctions couple to
give "minibands"

Miniband cascade laser (Superlattice Design)

